In [1]:
```
import os
os.chdir('C:/Users/User/Desktop/Results')
print(os.getcwd())
```

```
C:\Users\User\Desktop\Results
```

In [2]:
```
from __future__ import print_function, unicode_literals, absolute_import, division
import sys
import numpy as np
import matplotlib
matplotlib.rcParams["image.interpolation"] = 'none'
import matplotlib.pyplot as plt
%matplotlib inline
%config InlineBackend.figure_format = 'retina'

from glob import glob
from tifffile import imread
from csbdeep.utils import Path, normalize
from csbdeep.io import save_tiff_imagej_compatible

from stardist import random_label_cmap, _draw_polygons, export_imagej_rois
from stardist.models import StarDist2D

np.random.seed(6)
lbl_cmap = random_label_cmap()
```

load in the model for prediction (from step 2)

In [3]:
```
    model = StarDist2D(None, name='stardist_2', basedir='labeling/stardist_model_2'
```

```
Loading network weights from 'weights_best.h5'.
Loading thresholds from 'thresholds.json'.
Using default values: prob_thresh=0.497334, nms_thresh=0.3.
```

define the folder of images to be labeled - input folder contains full size images of ovaries - construct output folder for the results

In [4]:
```
input_folder = "Fluorescent ovary study red only"
output_folder = input_folder+"_follicle_labels"
os.makedirs(output_folder, exist_ok=True)
```

define image axis normalisation

In [5]:
```
axis_norm = (0,1)
```

apply the model to all images within the defined input folder

In [6]:
```
# Iterate over each file in the input folder
for filename in os.listdir(input_folder):
    if filename.endswith(".tif"):  # Assuming all input images are TIFF files
        # Construct the full path of the input image
        input_image_path = os.path.join(input_folder, filename)
```

```
        # Load the input image
        img = imread(input_image_path)
        img = normalize(img, 1,99.8, axis=axis_norm)

        # Apply the model to predict labels for the image
        labels, details= model.predict_instances(img)

        # Construct the full path for the output file
        output_image_path = os.path.join(output_folder, filename)

        # Save the image labels
        export_imagej_rois(output_image_path+'.zip', details['coord'])

print("labeling complete.")
```

```
labeling complete.
```